Optimizing sampling and monitoring of species interactions within Biodiversity Observation Networks

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Abstract: Optimal monitoring strategies should be designed to efficiently monitor all essential facets of biodiversity. Yet, species interactions are often overlooked in monitoring designs compared to spatial coverage and species richness, partly due to the inherent difficulty of sampling and monitoring interactions compared to species distributions. Here, we used simulations to test the efficiency of monitoring species interactions and species richness within Biodiversity Observations Networks (BONs). We tested several methods for designing and optimizing BONs to monitor species interactions and examined efficiency when increasing the number of monitored sites. We showed that the required sampling effort is considerably higher for interactions, especially when considering that interaction realization and detection realistically depend on species abundance. Thus, expectations to monitor a relevant proportion of interactions within a BON should be lower than expectations to monitor its species. However, from a single-species perspective, optimizing monitoring designs based on a known species range proved an efficient strategy to retrieve its interactions, outperforming designs based on total species or interaction richness. Only optimizations based on detailed knowledge of where realized interactions occurred allowed bet`ter performance, highlighting the need for better knowledge of where interactions are likely to take place and integration of factors influencing interaction realization, such as species abundances, for further efficiency gains. Therefore, our results offer insights into defining realistic expectations for interaction monitoring. With the right target and optimization strategy, available information, such as species ranges, can guide observation network designs to monitor species interactions as central biodiversity components.

Keywords: biodiversity monitoring, species interaction networks, biodiversity observation networks, sampling design

Introduction

Biodiversity monitoring should encompass all facets of biodiversity. Article 7 of the Convention on Biological Diversity (United Nations 1992) calls for Parties to "[i]dentify components of biological diversity important for its conservation and sustainable use", and "[m]onitor, through sampling and other techniques, the components of biological diversity". Biodiversity Observation Networks (BONs) are promising initiatives to improve monitoring across scales and biodiversity facets. BONs are coordinated monitoring networks designed to harmonize observation systems, assist reporting on international assessments, and reinforce scientific basis in biodiversity monitoring (Navarro et al. 2017; Gonzalez et al. 2023). BONs can be thematic (Marine BON, Freshwater BON), national (China BON, French BON, Colombia BON) or regional (Asia-Pacific, Europe); some represent networks of monitoring sites for systematic monitoring (China BON) while others represent biodiversity data hubs (French BON) (Navarro et al. 2017). BONs aim to monitor biodiversity change, informing on both current status and trends (Gonzalez et al. 2025). It has been suggested that BONs should eventually be interlinked into the Global Biodiversity Observation System (GBiOS), a globally connected network providing capacity to monitor biodiversity change (Gonzalez et al. 2023), and guide actions needed for targets and goals of the Kunming-Montreal Global Biodiversity framework (CBD 2022). Additional BONs are currently in the development or planification stages, notably in Europe (Kissling et al. 2024) and in Canada (Gonzalez et al. 2025).

Recent work highlighted the need for improved sampling designs in BONs. Current sampling designs over large scales are often too sparse, at too coarse a scale, too infrequent, and without temporal replication (Santana et al. 2025). Recommendations for more effective spatial sampling design in Europe included stratified random sampling (with stratification across environmental, anthropogenic and political layers), incorporation of existing monitoring sites, filling of spatial gaps, co-location of monitoring activities (Kissling et al. 2024). Ecological monitoring design relies on site selection algorithms more often discussed regarding their capacity to achieve spatial balance and representative samples, and less frequently with an evaluation to performance to achieve an ecological monitoring goal (Norman and Poisot 2025). Species richness is the most commonly used biodiversity measure in monitoring contexts (Hillebrand et al. 2018), including in discussions over sampling designs. Maximising species richness can be an efficient strategy to design conservation networks while also capturing functional and phylogenetic diversity (Willig et al. 2023). However, the use of species richness to measure biodiversity change and guide biological conservation has been criticized, mainly since trends in species richness do not properly capture local changes through time as diagnostic indicators should (Hillebrand et al. 2018, 2025), while also neglecting species identities and not providing information on persistence of biodiversity and ecosystem services (Aguiar et al. 2024; Fletcher Jr. et al. 2025).

Species interactions are known to be a central component of biodiversity (McCann 2007; Valiente-Banuet et al. 2015; Harvey et al. 2017) , with their own spatial and macro-ecological dynamics (Baiser et al. 2019; Mestre et al. 2022; Windsor et al. 2023; Dansereau et al. 2024). We now have sufficient conceptual tools to better appreciate the value added that species interactions provide to efficient management actions (Dansereau et al. 2025; Moracho et al. 2025). However, species interactions are still considered to undergo a biodiversity shortfall, due to a lack of data at global scales (Hortal et al. 2015) as well as a fragmentation of type of interactions for which data are available across locations and environments (Poisot et al. 2021). Species interactions represent a critical component of ecosystem functioning and health, yet may be lost faster than species (Valiente-Banuet et al. 2015). Monitoring interactions is more informative regarding ecosystem functions than monitoring species richness or using EBVs (Jordano 2016). Network responses may be different from species richness (e.g. Doré et al. 2021), emphasizing the need for ecological monitoring to consider interactions, not just species richness. Species interactions can also improve our ability to predict responses to climate change (Abrego et al. 2021). For both these reasons, sampling designs should incorporate considerations for species interactions.

Prior investigations of interaction sampling do not address how efficiently we should expect to monitor interactions within the BON framework, nor did studies on the effectiveness of BONs pay particular attention to the sampling of species interactions. Much focus has been given on how to assess sampling completeness (e.g. Chacoff et al. 2012) and how sampling effort affects network properties (e.g., McLeod et al. 2021). Recent empirical (Caron et al. 2024) and theoretical (Poisot 2023) results have established that knowing the network structure does not always allow to know the list of interactions (and vice versa), therefore justifying the need to sample interactions as their own biological entities. A central challenge with the sampling of interactions is that it can be much less than sampling for species. Sampling effort required to document interactions is much higher than for species (e.g. 500% vs 64% in Chacoff et al. 2012). Evaluating sampling completeness requires assessing the proportion of forbidden links due to life history restrictions, which itself requires natural history knowledge and an analysis of sampling robustness (Jordano 2016). Some quantitative network properties such as connectance are less affected by sampling completeness than network composition or binary measure (Poisot et al. 2012; e.g. number of links, Vizentin-Bugoni et al. 2016). Small networks are also more sensitive to undersampling than larger ones (Henriksen et al. 2019). Quantitive frameworks exist to evaluate network taxonomic, functional and phylogenetic diversity from incomplete samples (Chiu et al. 2023).

A much needed element is to consider sampling network and interactions from an explicitly spatial perspective with variation in local structure—that is, measuring sampling effort from specific sites where interactions are sampled. For instance, Poisot et al. (2012) measured sampling effort based on resampling a dataset to document its effect on network composition and properties. Other examples focus on scale, notably showing that many network structure metrics are robust across spatial scales when accounting for expected increases of richness and decreases in connectance (Wood et al. 2015). In contrast, McLeod et al. (2021) used data from specific lakes compare sampling strategies, showing that sampling larger to smaller lakes better captures regional metaweb properties. We now have frameworks and models to describe and investigate the variation and detection interaction networks in space and time (Poisot et al. 2015; Cirtwill et al. 2019; Catchen et al. 2023), and therefore the next is to consider how to optimize monitoring for species interactions. Simulations have already in the been used in the context of optimizing monitoring to: generate communities and evaluate the potential of metrics to track biodiversity change (Santini et al. 2017), simulate networks and sampling to evaluate effects on estimated metrics compared to a known reference (de Aguiar et al. 2019, although not in a spatial context), determine optimal sampling frequency for monitoring (Daugaard et al. 2024), to examine the dependence of monitoring efficiency and trends detection given the number of monitoring sites, species abundance, detection probability, strength of decline, under a range of scenarios and monitoring schemes (Ficetola et al. 2018).

In this manuscript, we address three key questions. First, how should we implement monitoring programs across space to efficiently capture network composition and changes, as essential components of biodiversity? Second, can we monitor species interactions as efficiently as species ranges? Finally, what would be realistic expectations for the effectiveness of sampling designs to monitor species interactions?

We use spatially explicit community simulations to test the efficiency of monitoring species interactions within BONs, and explore various algorithmic and design options for network optimization. We aim to define realistic expectations for monitoring species interactions, in comparison to species richness and species ranges. Using simulation studies, we compare design and optimization options, examining how to assess and compare their efficiencies. We conclude with recommendations on how this should influence the design of BONs to improve the monitoring of species interactions as an important biodiversity component.

Methods

We used simulations to generate species interaction networks, species ranges, and biodiversity observations networks (BONs) with the goal of evaluating monitoring potential for interaction networks within BONs. We first detail the steps for the general model we used (Figure 1). We then go over three simulation studies we performed to investigate specific elements related to the monitoring of species interactions.

General model

We followed the approach and process-based, spatio-temporal model introduced by (Catchen et al. 2023; building on Cirtwill et al. 2019) to simulate interaction realization and detection while accounting for spatial variation in species occurrences, implemented in SpeciesInteractionSamplers.jl. This model broadly captures the different processes involved into turning a *potential* interaction (which is biologically feasible) into a *realized* one (it took place at a specified location), as outlined in Morales-Castilla et al. (2015). To the generative model originally presented, we add the design of Biodiversity Observation Networks and the evaluation of sampling efficiency between and across simulations for species interactions (Figure 1A).

Step 1: Generating interaction networks with spatial variation

First, we simulate networks with variation in species and interaction composition across a simulated landscape (Figure 1A, Step 1). To do so, we generate autocorrelated ranges representing the distribution of species across the landscape using neutral landscape models [NLMs; Gardner et al. (1987); Etherington et al. (2015)]. NLMs are used to generate landscapes replicating realistic autocorrelated distributions without underlying biotic or abiotic factors, and therefore represent useful baselines to explore ecological processes in a spatial context (Hesselbarth et al. 2024). NLMs have been used to generate realistic landscapes in contexts relating to biodiversity distribution and preservation, for instance to compare avian diversity between conservation approaches in coffee-growing landscapes (Valente et al. 2022) or to assess the potential of a suggested umbrella species in an ecotone (Duchardt et al. 2023). NLMs have also been used with species interactions, for example to evaluate the effect of landscape

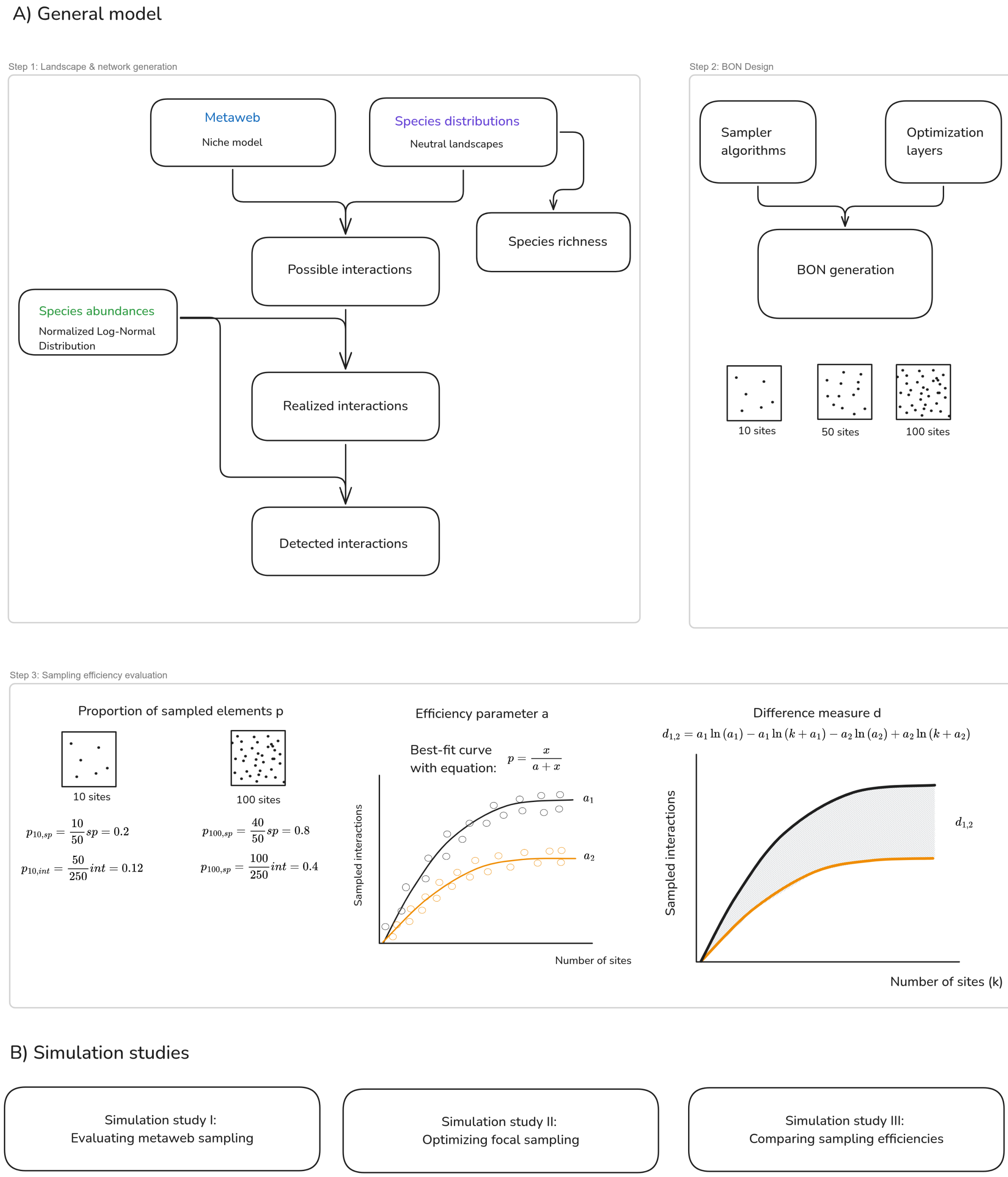

Figure 1: Conceptual diagram of the workflow and simulations. The general model (A) was used for the simulations, starting from generating landscapes with spatial variation in species and interaction composition, designing biodiversity observation networks (BONs) and measuring sampling efficiency within and across simulations. The simulation studies (B) use the general model to investigate specific elements related to sampling interactions with BONs.

structure in cross-species disease transmission between interacting species (Forero-Muñoz et al. 2025). Here, we use NLMs to create communities of known composition and species richness at every location in our simulated landscape.

Next, using the spatio-temporal model presented by Catchen et al. (2023), we define the **feasible interactions** between the species by generating a metaweb using the niche model (Williams and Martinez 2000). The model defines interactions by assigning species a niche position and feeding range, similarly to body-size relationships between predator and preys, and generates networks with structural properties similar to empirical food webs (Williams and Martinez 2000; Gravel et al. 2013; Delmas et al. 2019). It is widely used as a generative model, including in the context of investigating interaction sampling (Poisot et al. 2012; Wood et al. 2015).

Combining the species ranges and the metaweb, we verify where species with feasible interactions co-occur, which returns the list of **possible interactions** (Catchen et al. 2023) at every location in our landscape. Next, to define **realized interactions**, we incorporate a consideration for species abundances, which influence both interaction realization and detection. For example, rare and low-abundance species are less likely to interact in a location because they are less likely to encounter one another, especially compared to abundant species, leading to neutrally forbidden links (Canard et al. 2012; Catchen et al. 2023). The neutral model introduced by Catchen et al. (2023) first generates a distribution of species relative abundances from a log-normal distribution, then models interaction realization as a stochastic Poisson Process, drawing realized interactions from a Poisson distribution parameterized on the relatives abundances of interacting species and a fixed realization energy parameter (see Catchen et al. 2023 for full details).

Finally, we define the set of **detected interactions** by also conditioning the detection of realized interactions on the species abundances (even if an interaction is realized in a location, we need to be there at that moment to detect, which is again less likely for low probability events; see Catchen et al. (2023) for implementation). The result of this model is a landscape with spatial variation in species and interaction composition and detection, where species composition, realized interactions, and detected interactions are based on the process model and known at all locations, allowing to compare the efficiency of sampling strategies.

Step 2: Designing Biodiversity Observation Networks

We design Biodiversity Observation Networks (BONs) across the simulated landscape and measure their efficiency at monitoring the generated interaction networks. Norman and Poisot (2025) presented an overview of the design and site selection process accounting for ecological monitoring goals. The design process uses a site selection algorithm to designate candidate sites for the monitoring network. Key features of the site selection algorithms highlighted by (Norman and Poisot 2025) include the ability to balance sites across space for equal representation (see Benedetti et al. 2017; Kermorvant et al. 2019 for reviews), to stratify across regions or populations, to include auxiliary site information (environmental or biological) to weight inclusion probability, and to produce master samples including current monitoring site or from which to select a subset of sites for a specific monitoring project (van Dam-Bates et al. 2018; Robertson et al. 2024).

Using our model simulations, we can test different algorithms to distribute monitoring sites across the landscape and optimize their location given different information such as species ranges, species richness or presence of realized interactions. For example, we can use the Balanced Acceptance Sampling algorithm (Robertson et al. 2013) to design BONs when aiming to balance sites across our simulated landscapes, mostly for its efficiency and its ability to weight site inclusion probabilities based on a provided layer. On the other hand, to optimize given available information, we can use an algorithm such as Uncertainty Sampling (Lewis and Gale 1994), an active learning method from the machine learning field aiming to improve a model by targeting its least certain elements (Settles 2009). Instead of focusing on uncertainty, we can use this concept to target the sites with the highest values in a layer provided for optimization, with the idea that it could lead to the highest information gain for the BON. Examples of active learning relating to biodiversity monitoring and sampling site selection include improving species range estimation from a smaller number of actively selected sites (Lange et al. 2023), minimizing field data collection costs based on distances between samples (Malek et al. 2019), and identifying uncertain labels for classification to improve bioacoustic monitoring performance across stratified samples (McEwen et al. 2025).

Step 3: Evaluating sampling efficiency with comparable measures

Our simulations require evaluating sampling efficiency at three levels (highlighted in Figure 1, Step 3) : at a landscape scale for a given BON and number of sites (a snapshot description measure), as an efficiency curve describing the efficiency with an increasing number of sites (a snapshot ensemble summary measure), and as a difference score to compare optimization options (an ensemble comparison measure).

On the first level, we use the **proportion of sampled elements** (species or interactions, noted p) to evaluate how efficient a given BON is on a landscape scale. In simulated landscapes, species and interaction composition is known everywhere in our simulated landscapes; thus we can determine what can be sampled for each generated BON (e.g. one made of 50 spatially-balanced sites). We can then evaluate their efficiencies as the proportion of sampled elements compared to the true richness (of species or interactions), again known in the simulations. This can be seen as a snapshot view of a single BON efficiency.

On the second level, we define an **efficiency parameter** (a) describing how the sampling efficiency changes as we increase the number of sites in a BON. Ideally, we need one summary measure to represent this ensemble of snapshots and summarize the potential BON-efficiency for a given landscape and optimization strategies. Moreover, this measure needs to be comparable across samplers, optimization layers and species degrees, given how the generated metawebs and landscapes might vary between independent simulations. Since the exact proportion of monitored interactions is known in our simulations, we used the following single-parameter equation which generates similar efficiency curves:

$$p = \frac{x}{a + x} \tag{1}$$

where p is the proportion of monitored interactions, x is the number of sites in the BON and a is the efficiency parameter we use for comparison. Here, a is the single parameter describing the shape of the curve, and can be compared across samplers, optimization layers or focal species degree. The lower this parameter, the faster we reach a high proportion of monitored interactions. We can identify the value of a yielding the curve with the best fit to the observed data, as described in our Simulation study III.

On the third level, we define a **difference measure** (d) to compare optimization algorithms and layers for the same simulated landscape. We have efficiency curve parameters a, which we can compare between options on a greater or lower basis, but the order of magnitude of the difference is not necessarily comparable across sets of simulations, hence the need for a more formal measure. To compare the efficiency curves within a given simulation (i.e., between samplers and optimization layers for a given metaweb and set of species ranges), we used the definite integral of Equation 1 between 0 and k, the maximum number of sampling sites in the landscape (10,000), which takes the form:

$$\int_0^k \frac{x}{a + x} \, \mathrm{d}x = a \ln(a) - a \ln(k + a) + k \tag{2}$$

Using Equation 2, we can compute the difference in efficiency between two curves as the difference of integrals:

$$d_{1,2} = a_1 \ln(a_1) - a_1 \ln(k + a_1) - a_2 \ln(a_2) + a_2 \ln(k + a_2) \tag{3}$$

This difference measure allows to identify which samplers or optimization layer is more efficient, quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs. We can then compare all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges).

Simulation studies

We performed three simulation studies to investigate the efficiency of monitoring species interactions within BONs. Across our simulations, we used landscapes of 100 x 100 pixels, 75 species, and a connectance of 0.2 to generate metawebs using the niche model. The connectance and number of species are within the range of values of well-characterized empirical food webs (Williams and Martinez 2000; Dunne et al. 2002, 2004; Smith-Ramesh et al. 2017) commonly used for reference (Williams and Martinez 2008; Baiser et al. 2010; Wood et al. 2015). These parameters are also within the range of previous simulation studies on spatial ecological network dynamics, *e.g.* Thompson and Gonzalez (2017). Exploratory simulations showed the number of sites in the BON to be the main element affecting the proportion of sampled species or interactions compared to other model parameters such as metaweb connectance and landscape autocorrelation. The activation energy parameter expectedly influenced results as covered in Catchen et al. (2023) when presenting the model.

Simulation study I: Evaluating metaweb sampling with spatial balance compared to species richness

Our first simulation study aimed to investigate the efficiency of sampling networks across space to document a metaweb of interactions, comparing it with the efficiency to document species richness. We generated spatially balanced Biodiversity Observation Networks (BONs) across our simulated landscape, defining sites where to sample species and interactions, and testing all number of sites between 1 and 100. We then evaluated the sampling efficiency of each BON in terms of overall species richness and compared to the generated metaweb for each interaction type. For species richness, we assumed we sampled all species present in a sampled location, combined this information across all sampled sites to determine the number of species encountered through sampling, and compared the result with the total species richness in the landscape (75 species). For interactions, we aggregated each interaction type separately (possible, realized and detected) for a given BON and evaluated the proportion of the metaweb recovered through sampling (and therefore for a given number of sites). We ran simulations with 100 fully-independent replicates (different metaweb, species ranges, abundance distribution, realized interactions) for each number of site in a BON, yielding 10,000 simulations in total.

Simulation study II: Optimizing a focal sampling on a single-species

Next, we investigated sampling efficiency using a focal species perspective. We simulated a focal sampling where an observer would target one species and aim to retrieve all of its interactions (here known from the metaweb), but might have different information to optimize its sampling. Observation network design and efficiency needs to be evaluated beyond spatial balance, and also regarding their efficiency at meeting biodiversity monitoring objectives (Norman and Poisot 2025). Thus, the idea of our focal sampling exploration is to represent a simplified case with a clear objective (retrieving a species' interactions) and where we do have prior knowledge about the system, either through empirical data or model predictions, which can be used to optimize sampling.

Optimizing focal sampling relies on two elements: the sampler algorithm used and the information provided for optimization. Species range represents an attainable level of information an interested observer could have, for instance based on expert knowledge or species distribution models. Therefore, we assumed the observer knew the species range across the landscape (presence or absence). We tested three sampling algorithms for comparison: simple random sampling (to represent uninformed sampling), weighted balanced acceptance sampling (aiming to balance sites across space while weighting favorably for species ranges) and uncertainty sampling (which targets locations with a higher value in the layer provided). Next, we tested four information layers for optimization: one with the focal species range (our attainable information knowledge for the observer), one with the probabilistic range (for a less informed case, similar to the output of a species distribution model), one with the location of realized interactions for the focal species (to represent a best-case scenario) and one with species richness across the landscape (to represent a holistic level of information for the system).

Using the approach described in the previous section, we generated a set of metaweb, species ranges, and locations for realized interactions (to showcase a specific example, and counter-verified with separate sets which yielded conceptually similar results, as described in the next section). We selected the species with the highest degree (number of interactions) in the metaweb as the focal species for the simulation. We then generated BONs of 1 to 500 sites (at interval increments of 5 sites) using the three possible algorithms (simple random, weighted balanced acceptance, uncertainty sampling) and four optimization layers (focal species range, probabilistic range, realized interactions, species richness), with 100 replicates for every number of sites. For each BON generated, we extracted the number of unique realized interactions sampled across all sites (the monitored degree for the focal species).

Simulation study III: Comparing sampling efficiency across simulations

We verified whether our results were constant when replicating simulations with different simulated metawebs and species ranges. We ran 100 independent simulations, each with 50 replicates for every number of sites (once again from 1 to 500 sites at interval increments of 5 sites). Doing so raised the issue of describing and comparing the efficiency of the simulated monitoring across samplers, optimization layers or focal species degree, as generated metawebs and landscapes varying between independent simulations, hence the need for the comparison measures described earlier in Equation 1 and Equation 3. We used the efficiency parameter a (Equation 1) to describe efficiencies between independent simulations and the difference measure d (Equation 3) to compare efficiencies of optimization options within a specific landscape.

For each independent simulation, sampler and optimization layer option, we performed a grid search across possible values of a (from Equation 1) to identify the curve yielding the best fit with the simulation's result. We searched for parameter values on an exponential scale as it allowed for more gradual control over the curve shape, evaluating 10,000 evenly spaced values between e^{-5} and e^{15} (limits chosen to go beyond the fitted values), and selected the value of a yielding the curve minimizing the sum of squared errors to the simulation results.

We compared all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges), using our difference measure d (Equation 3). The measure allows to identify which samplers or optimization layer is more efficient, thus quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs.

Results

Simulation study I: Lower sampling expectations for species interactions than species richness

Our results for metaweb sampling highlight a marked difference between the proportions of sampled species and interactions (Figure 2). Sampled species saturated very quickly, with all species sampled with less than 25 sites in a BON across the landscape. Possible interactions, which depend on co-occurrence of species with a feasible interaction in the metaweb, also saturated quickly, reaching over 90% of interactions in the metaweb with 25 sites and increasing asymptotically afterwards. In contrast, realized and detected interactions, which account for the impact of species abundances, reach much lower proportions, both under 25% of the metaweb after 100 sites. While the proportion would keep increasing with a higher

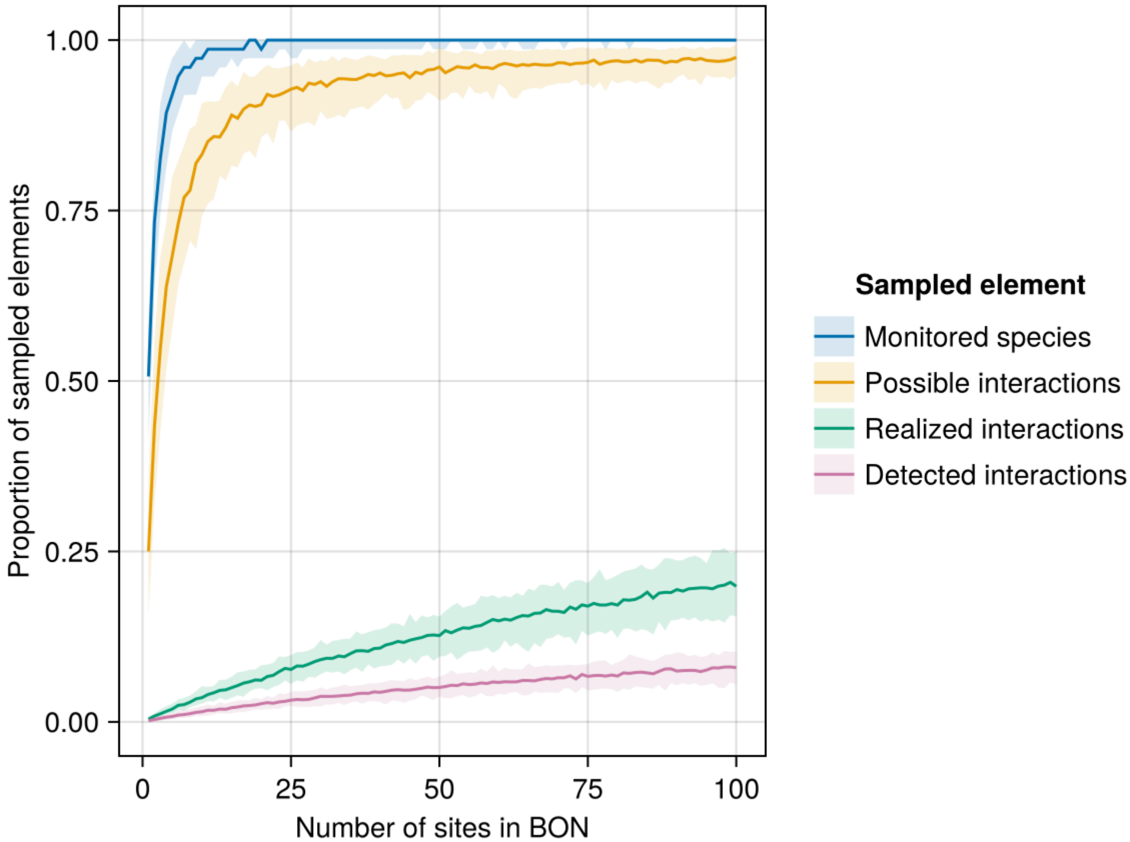

Figure 2: Proportion of sampled species and interactions for a given number of sampling sites in a Biodiversity Observation Network (BON) across the simulated landscape. The proportion refers to the total species richness and the total number of interactions in the metaweb as the maximum reference. The central line indicates indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates (100 per site, 10,000 in total).

number of sites, the speed at which interactions accumulate is notably lower than for species richness and possible interactions, highlighting the need for a much higher expected sampling effort.

Simulation study II: Differences between samplers and optimization layers

Our focal sampling experiment showed the Uncertainty Sampling algorithm to be most efficient at optimizing BON-design based on a known species range and while aiming to maximize the sampling of interactions of a single focal species (Figure 3A). Simple Random sampling, as a proxy for uninformed sampling, performed the worst among the algorithms tested. Weighted Balanced Acceptance performed notably less than Uncertainty Sampling at maximizing the efficiency for the focal species, although it maintained a better spatial balance across the landscape.

Optimizing on the known species range, our proxy for an attainable level of information an interested user might have, was a more efficient strategy to retrieve the focal species' interactions than optimization based on species richness (or interaction richness), effectively sampling more than half of the species' interactions (Figure 3B). In contrast, optimization based on the location of realized interactions, our best-case scenario requiring a higher level information, expectedly performed better, retrieving most of the species' interactions.

Simulation study III: Comparison of efficiency parameters across simulations

The distribution of efficiency parameters across independent simulations confirmed the interpretations based on Figure 3, but also highlighted a wide range of outcomes in efficiency between simulations. Among samplers, the median efficiency parameter indicated a better performance (lower parameter value) for Uncertainty Sampling, followed by Weighted Balanced Acceptance and Simple Random (Figure 4A). The actual efficiency values spanned a similar range across the possible sampler options with all simulations combined. Among optimization layers, optimizing on the focal species range performed better than on the probabilistic range, followed by species richness, although all three options spanned similar ranges with all simulations combined (Figure 4B). In contrast, optimizing based on the realized interactions distinctly and consistently performed better across all simulations.

The fact that multiple options span similar ranges of efficiency values across independent simulations highlights the need to compare the efficiency within a given simulation. Comparing options one-by-one clearly highlighted a most efficient one in all case (Figure 5, Supp. Mat.), consistent with the showcased single-simulation example (Figure 3). Uncertainty Sampling was more efficient than Weighted Balanced Acceptance in 98 of 100 simulations, and both were more efficient than Simple Random in all simulations (Figure 5). Optimizing on the focal species range was always more efficient than on the probabilistic range or species richness (with probabilistic range more efficient than species richness in all but one simulation), but always less efficient optimizing on realized interactions.

While the simulations clearly highlight which sampler or optimization layer is most efficient across all simulations, efficiency parameters and differences vary over a wide range of values. We verified if this could be tied to the occupancy of the focal species in simulated landscape, measured as the number of cells where the species is present. We examined the relationship between occupancy and efficiency. Optimizing using Uncertainty Sampling on the focal species range and on realized interactions is especially efficient at low occupancy, as it targets the fewer cells where the focal species is present. As species occupancy increases, the difference between options decreases. When the focal species has high occupancy and is present almost everywhere across the landscape, Weighted Balanced Acceptance and Simple Random perform similarly to Uncertainty Sampling, as balancing sites across space

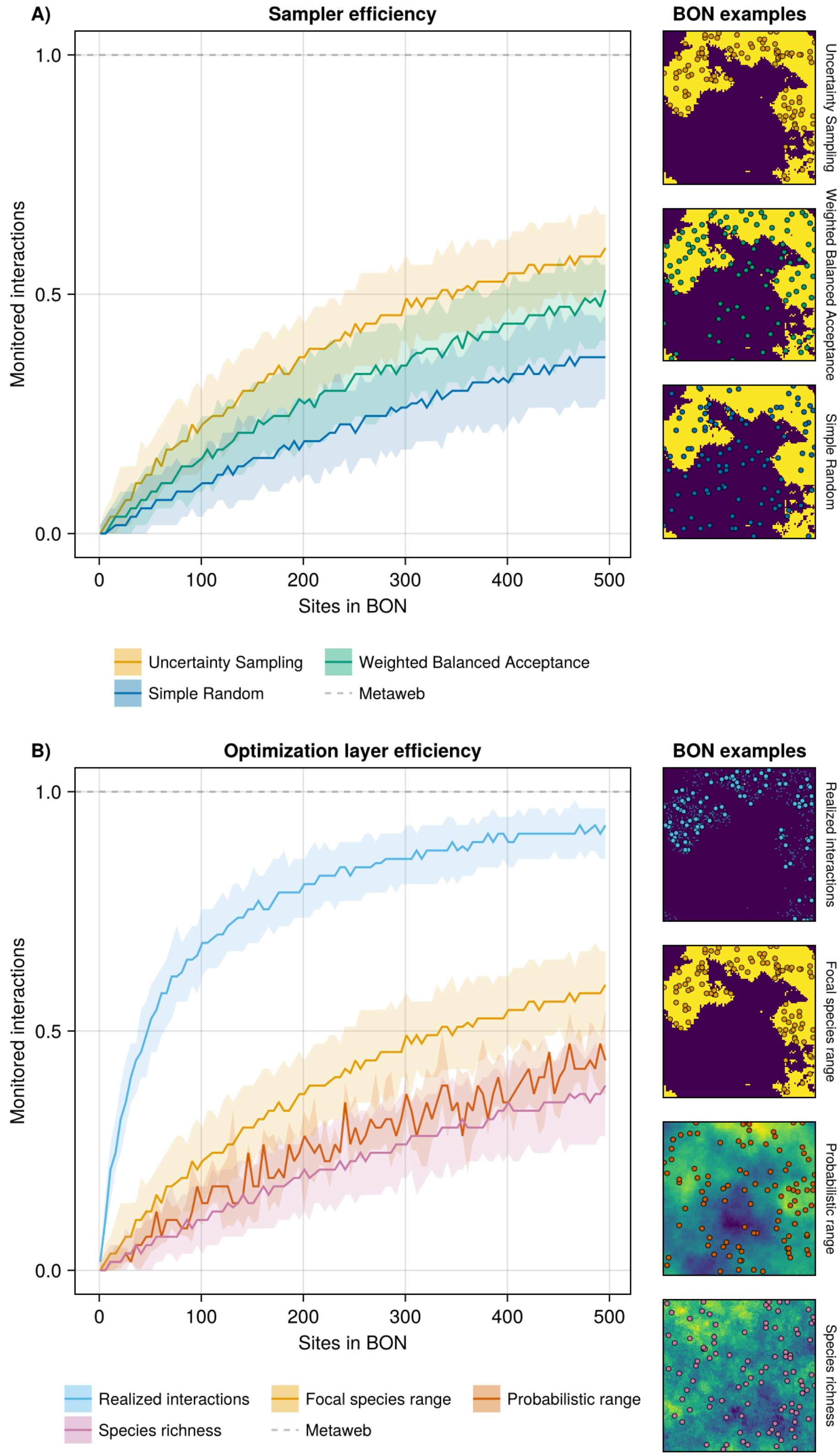

Figure 3: Efficiency of sampler algorithms (A) and optimization layers (B) at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). Panel A compares three samplers used with the species range as the optimization layer while Panel B compares four optimization layers with Uncertainty sampling as the sampler. The central line indicates indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the landscape (species present at pixels in yellow).

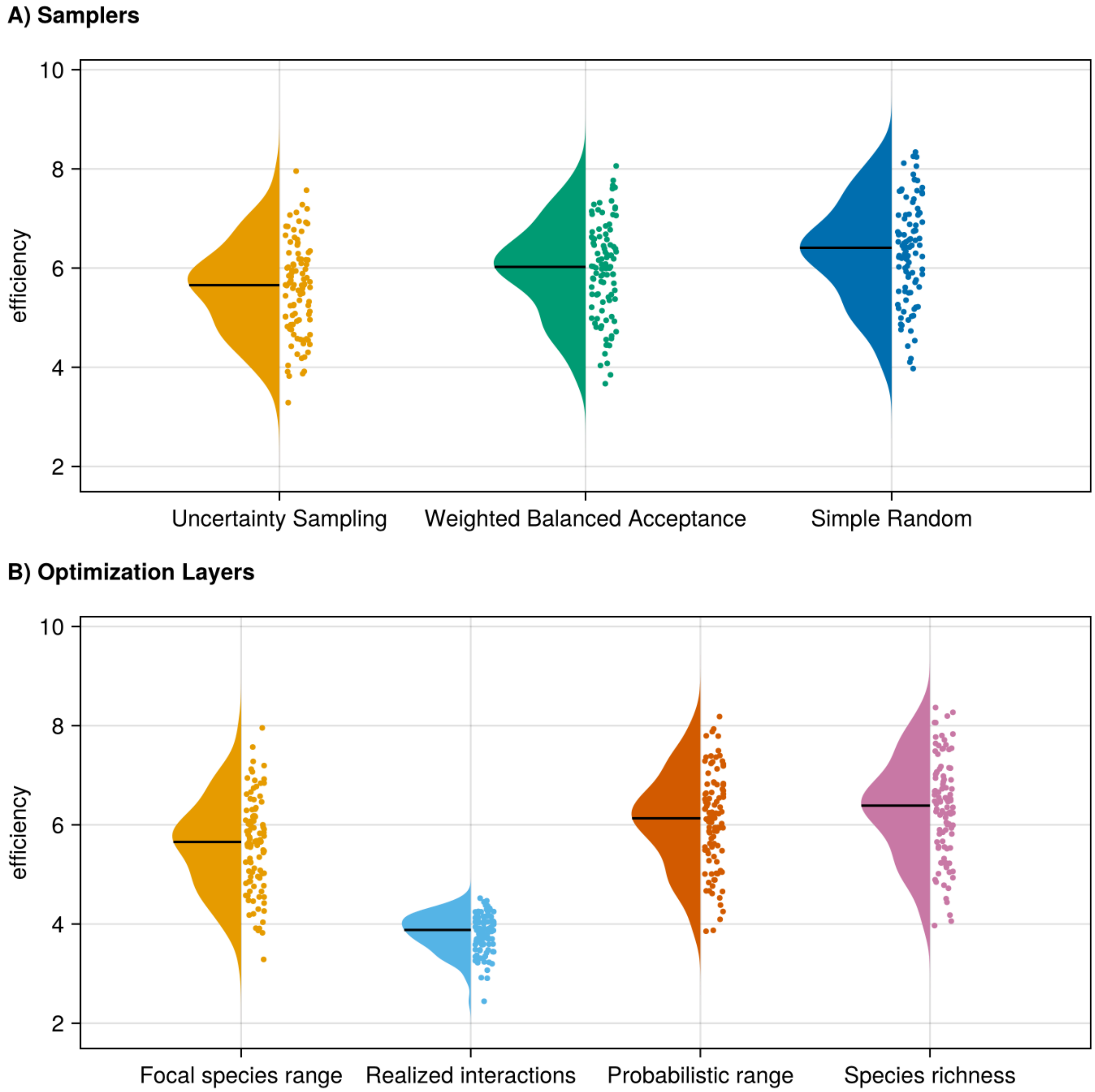

Figure 4: Distribution and density of efficiency parameters across 100 independent simulations for the three sampler options tested on the focal species range (A) and for the four optimization layer options using Uncertainty Sampling as the sampler (B). Uncertainty Sampling on panel A and Focal species range on panel B therefore represent the same data. The efficiency parameter is the natural log of parameter a in Equation 1, describing the best-fit sampling efficiency curve for a given simulation.

or choosing at random becomes equivalent to targeting the species range. Optimizing based on species richness became more efficient as occupancy increased. A possible explanation is that at high occupancy, species range is not a restricting element and thus targeting sites with high richness directly leads to targeting sites with more interaction partners. On the contrary, at low occupancy, species range is an important restriction, and optimizing on richness of all species might target sites where the focal species is absent. Optimizing on the probabilistic species range behave similarly to species richness; at high occupancy, it is very likely the

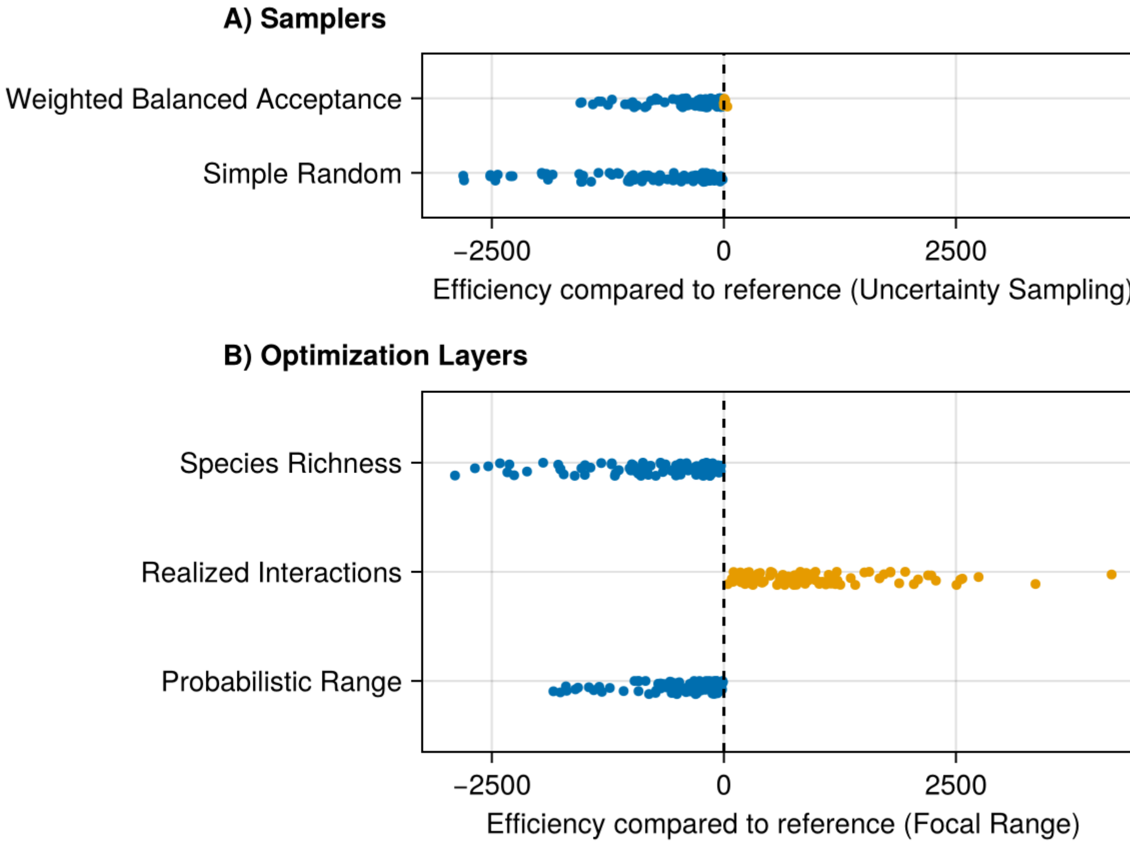

Figure 5: Within-simulation comparison of efficiency between samplers and optimization layers. The comparison values is based on the difference of the definite integrals of the efficiency samplers curves described in Equation 3. Negative values (in blue) indicate that the compared option lead to a less efficient sampling than its reference, while positive values (in orange) indicate it was more efficient. References were chosen for narrative purposes and to simplify the number of comparisons displayed. All one-by-one comparisons are available in Supp. Mat.

species will be present at the sites with high probabilities and therefore the sampler can more effectively filter out sites where the species is absent. At low occupancy, the focal species will be present in fewer locations with high probabilities, but might be absent from locations with intermediate probabilities, making it more challenging for the sampler to pick apart.

Discussion

What should be realistic expectations for sampling and monitoring of species interactions? Our simulations highlight they should be much lower than for species richness once considering that interaction realization and detection depend on species abundance. In other words, we can expect the monitoring of species interaction within a BON to require a (much) higher monitoring effort across space than the one required for monitoring species and community

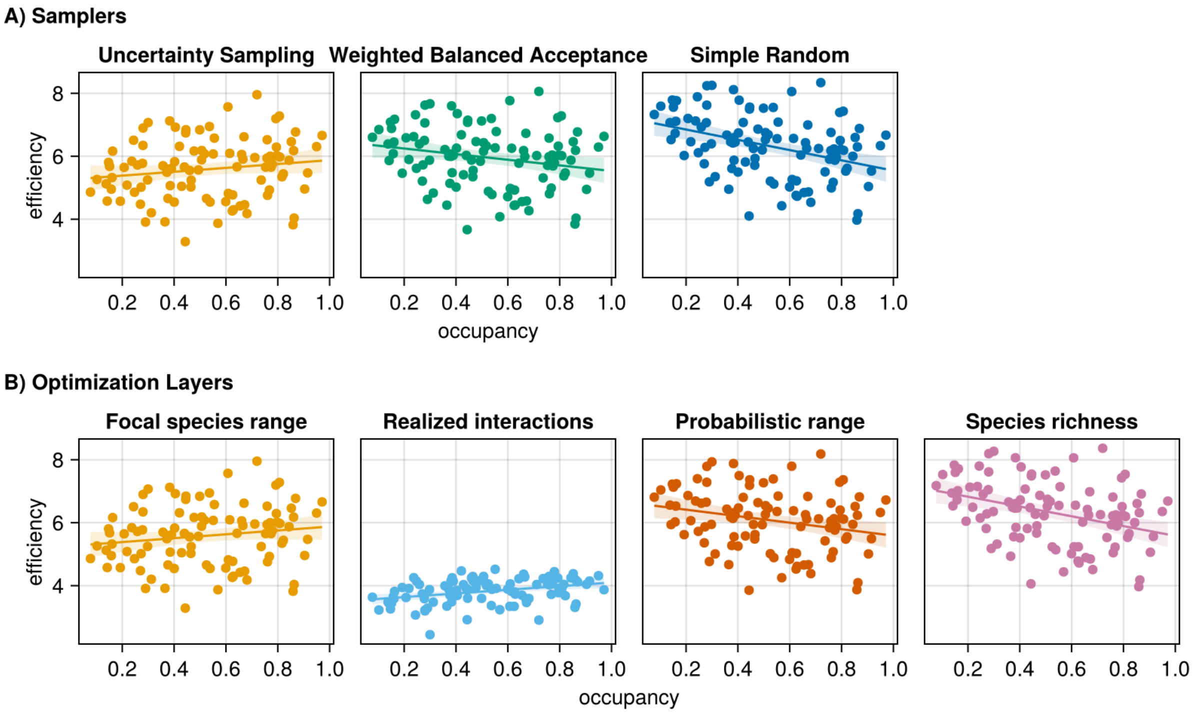

Figure 6: Relationship between the occupancy of the focal species and sampling efficiency for all optimization options. These results come from 100 independent simulations, and therefore 100 different sets of metawebs, species ranges, and occupancy of the focal species. The efficiency parameter is the natural log of parameter a in Equation 1, describing the best-fit sampling efficiency curve for a given simulation.

composition. This is similar to observations regarding sampling effort required to document pollinator networks (Chacoff et al. 2012), yet here framed in the context of BON design and monitoring across space .

We can improve coverage with the right optimization method, but efficiently doing so requires a clear and attainable sampling or monitoring objective. For instance, adopting a single-species perspective [*e.g.* focusing on interactions for a keystone or high degree species; Pires et al. (2017); de Aguiar et al. (2019)] can yield relevant results while relying on attainable levels of information. Optimizing on a known species range proved an efficient strategy to retrieve its interactions, better than optimization based on more holistic, community-level information such as species or interaction richness (identified in contrast by Willig et al. 2023 as an efficient target to maximize to conserve multiple biodiversity facets). Optimizing based on the location of the realized interactions was even more efficient. Yet, these simulations represent a best case scenario beyond reach in an applied context: it assumes that there is already pre-existing knowledge on where interactions have been observed, which is unlikely to exist for most interactions, regardless of their relevance for monitoring. Nonetheless, it highlights that the most important efficiency gains will come from better knowledge of where interactions are likely to take place, which suggests that monitoring of interactions will become more efficient over time. Better incorporating considerations for species abundance, the factor influencing interaction realization in our simulations, will be crucial and potentially more impactful than improving knowledge or reducing uncertainty over species ranges.

When does the choice of sampler or optimization layer matter the most? Here we concur with Norman and Poisot (2025) that ecologically-relevant metrics are required to evaluate the performance of monitoring designs, beyond spatial balance. However, while they found that many sampling algorithms performed similarly for biodiversity monitoring, our results highlighted better options when the monitoring target is highly localized. We observed the most important differences for species with low occupancy, where Uncertainty Sampling is relevant to target only the focal species range. Differences were less important at higher occupancy and for species present across the whole landscape, although Uncertainty Sampling still performed better. This result should be expected, as a specialized sampler should be more efficient when a species range is highly localized compared to one balancing sites across the the whole landscape. However, it also applies in broader context, highlighting how some monitoring targets (e.g., species interaction) expected to be highly localized or more likely in specific locations might benefit from targeted monitoring designs. In such case, efficiency gains will depend on improvements in our capacity to predict the location of those phenomena and on our ability to continuously evaluate and update predictions based on the information acquired.

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